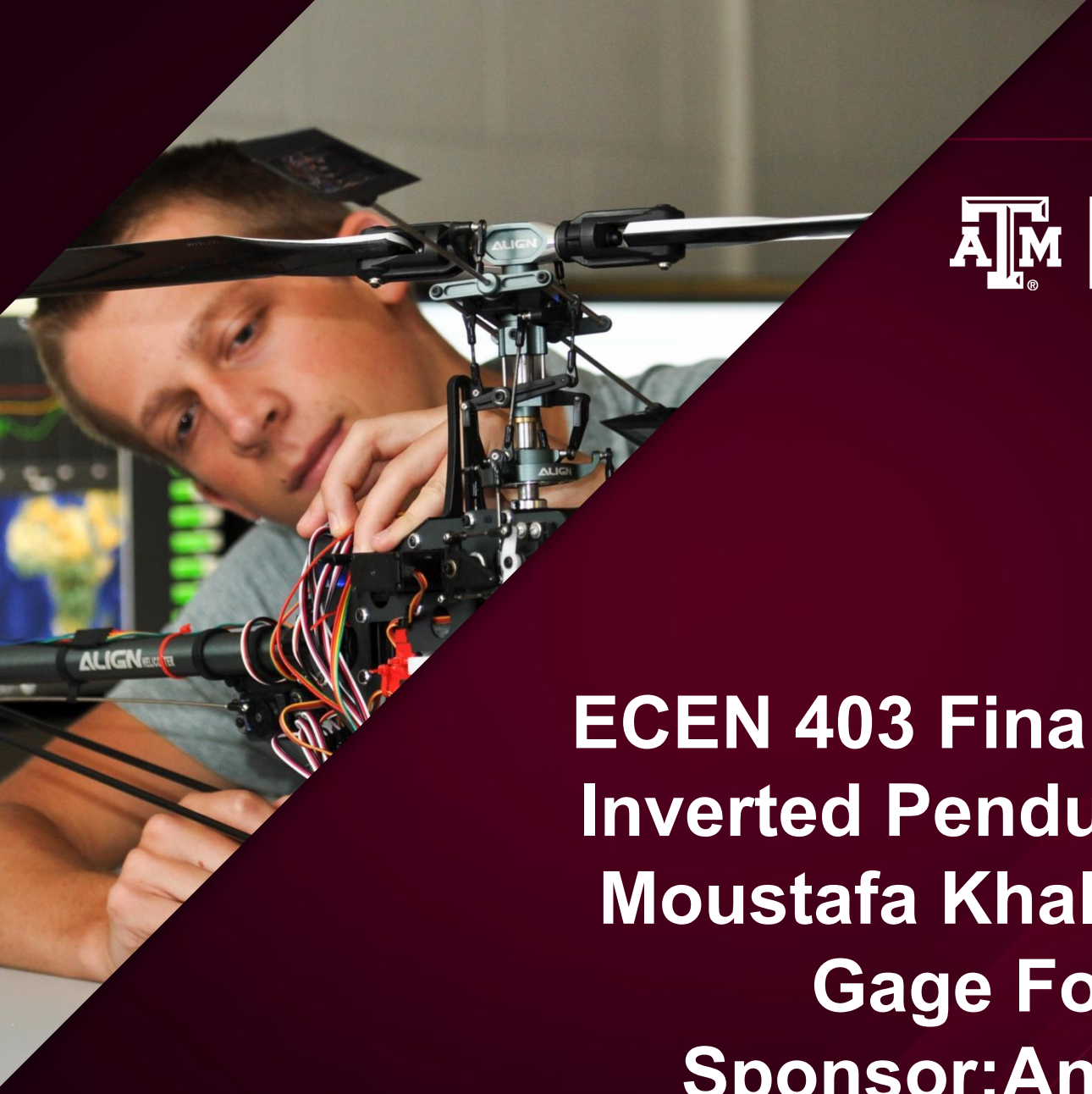

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ECEN 403 Final Presentation
Inverted Pendulum Robot #1

Moustafa Khalil, Kaitlynn Ly
Gage Fox, Eric Chang
Sponsor:Aniruddha Datta

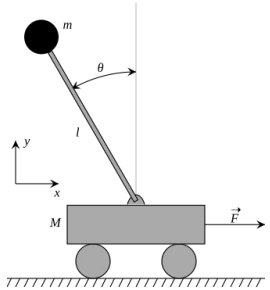
Project Summary

Problem statement:

- Autonomous warehouse robots generally experience instability when handling unbalanced loads or making sudden maneuvers. These movements can cause tilting, making balance a key challenge.

Our Solution Will:

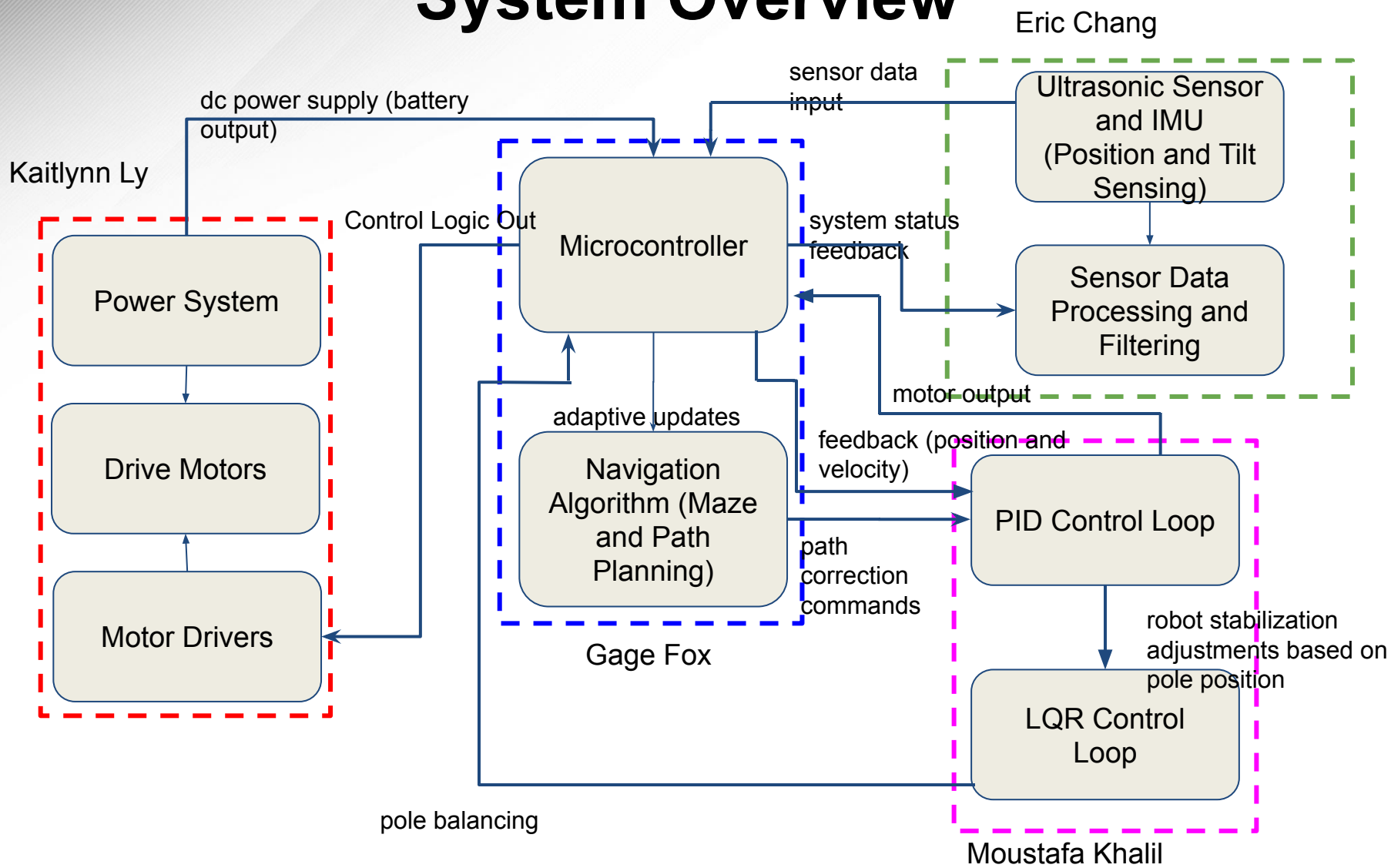
- Deliver an autonomous, self-navigating robot that maintains stability of an unstable payload while moving through constrained spaces.
- Integrate mechanical, sensing, power, and control subsystems into a cohesive, microcontroller-based platform designed for reliability and safety.



Side View

Isometric View

System Overview



Controls Subsystem - Moustafa Khalil

Completed Work

- Created model of the inverted pendulum
- Chose PiD and LQR controller
- Simulated the time response graph to verify stability of the controller

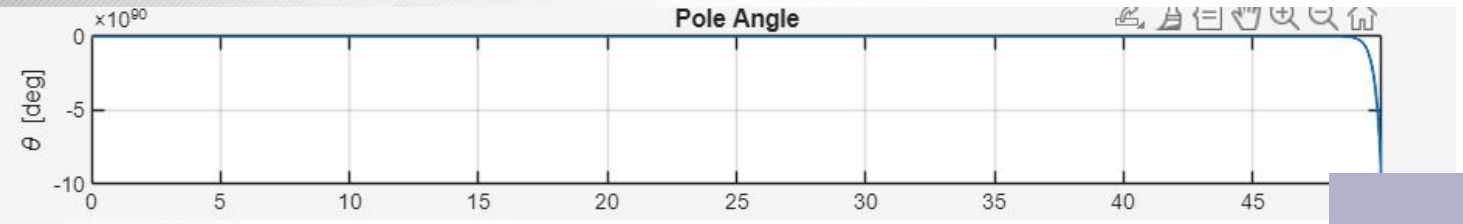
Continuing Work

- Validate the controller through a 3D model of the inverted pendulum
- Show pole angle graph over time of the simulation to confirm stability

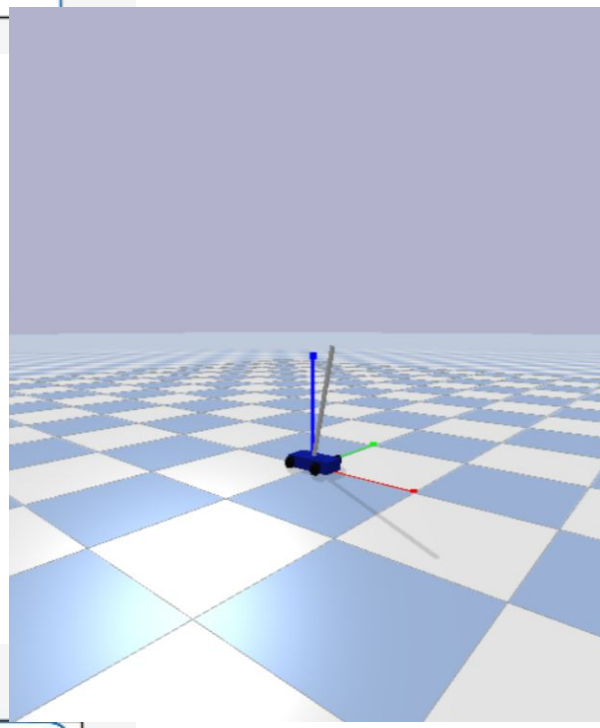
Planned for 404

- Tune the robot based on real physical characteristics
- Integrate with all the subsystems
- Validate controller

Controls Subsystem



Pole Angle x Time



Cart Position x Time

Power & Mechanical - Kaitlynn Ly

Completed Work

- Designed buck converter and LDO circuit schematic
- Designed and ordered voltage converter PCBs and its components
- Built robot chassis with motor mounts
- Developed overall power system connections integrating battery, protection, and regulation elements

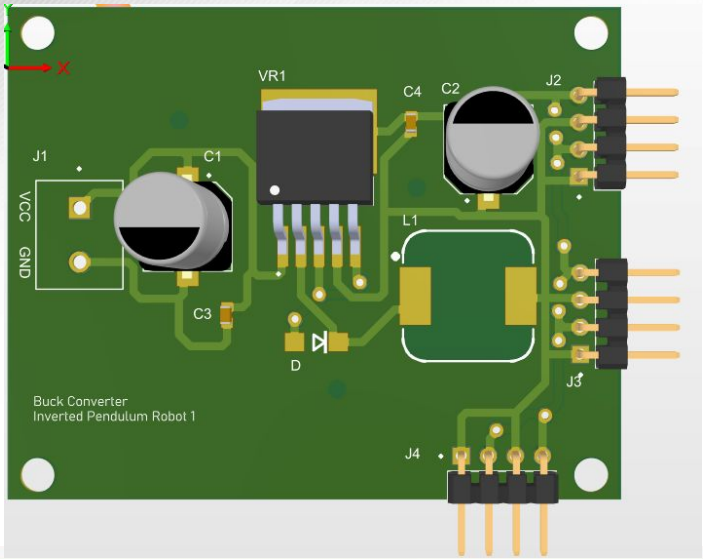
Continuing Work

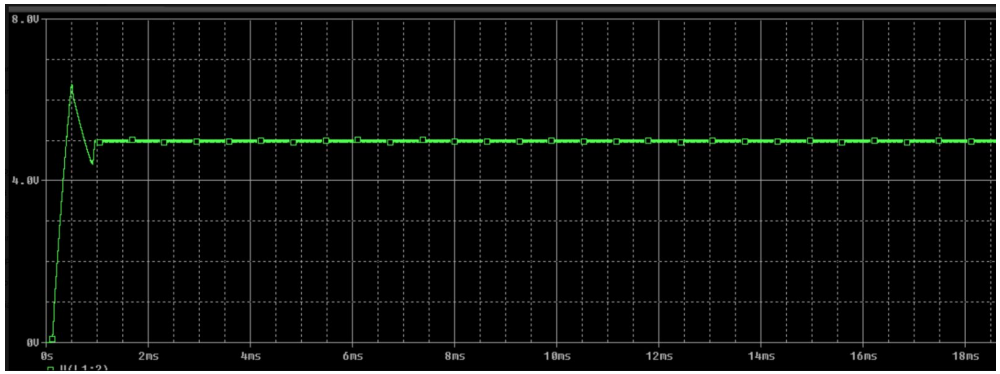
- Solder components onto voltage converter PCBs
- Still in process of validating protection elements of the system
- Pendulum assembly
- Full system integration

Planned for 404

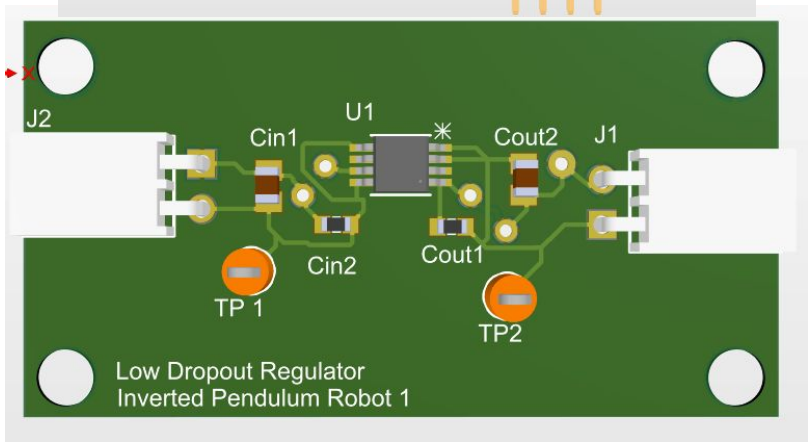
- Integrate with all subsystems to run properly
- Adjust robot chassis to mount all the electronic components
- Wiring design for the robot

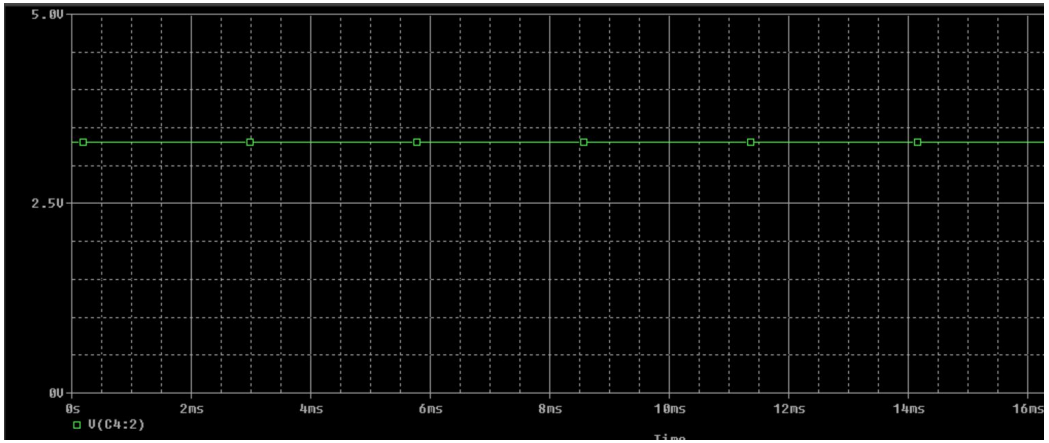
Power & Mechanical





5V Buck Converter Simulation with Load





3.3V LDO Simulation with Load

Microcontroller - Gage Fox

Completed Work

- Completed Main PCB Capture - Containing peripheral communications.
- Defined communication lines and links between STM32 and peripherals.
- Designed on-board programming circuit for STM32 using STLINK-V3MODS.

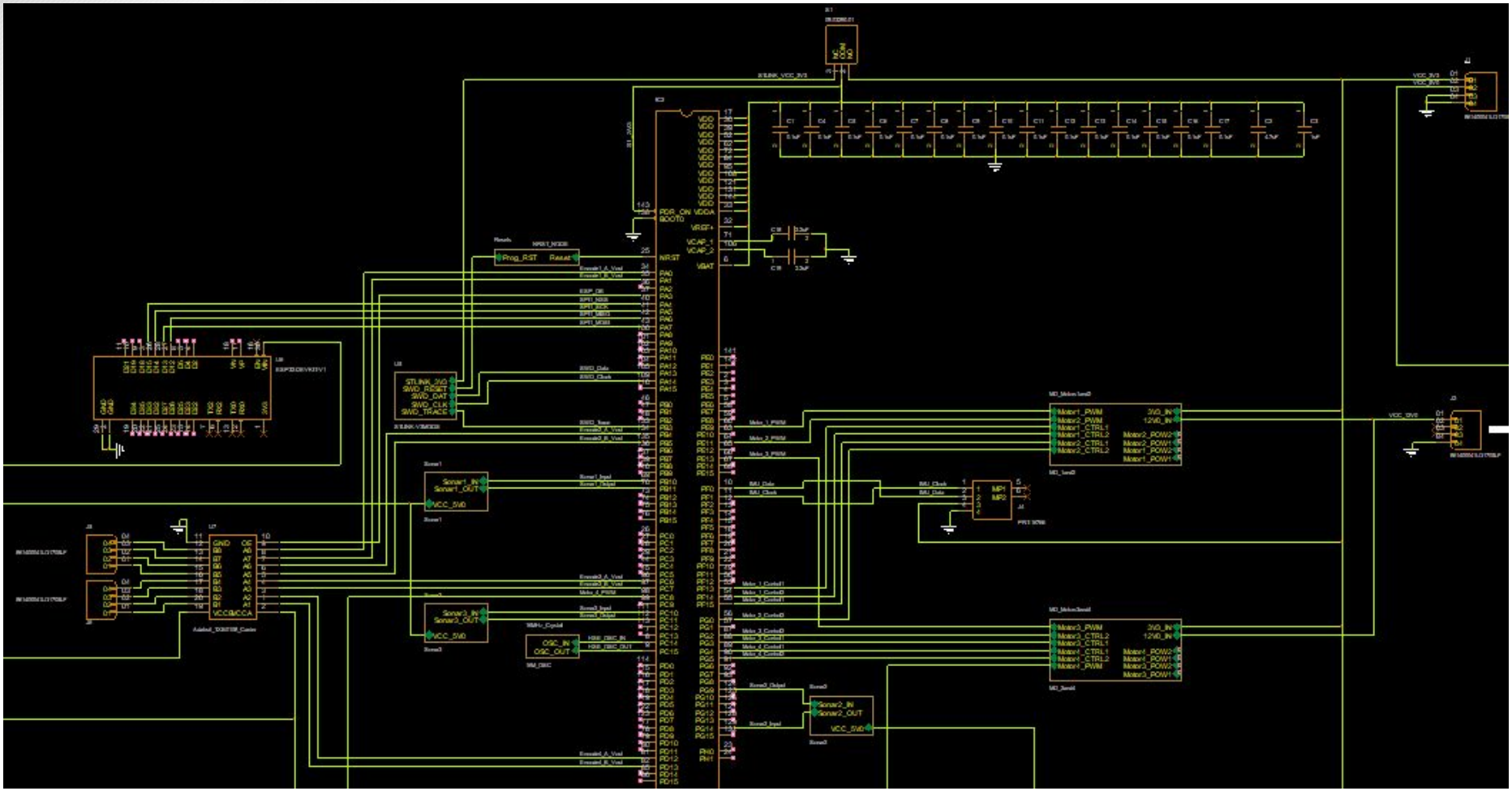
Continuing Work

- PCB Layout is still being implemented (expected completion within days).
- Coding SPI, UART, and I2C Communication protocols on STM32.
- Validating PCB design, communication reliability/speed, and data processing.

Planned for 404

- Coding control algorithms into STM32
- Integrating all peripherals with Logic PCB.
- Implementing sensor data communication to ESP32 wireless module as slave.
- Implementation of safety protocols/programming to prevent damage or harm.

Microcontroller



Sensors, Navigation, and GUI - Eric Chang

Completed Work

- Characterized ultrasonic sensor range and field of view; quantified IMU steady-state bias, drift, and noise
  - reliable detection ≈ ±10°
  - IMU steady-state orientation error ≈ ±0.0008°
- Developed ESP32-to-web telemetry for data visualization and diagnostics
- Implemented obstacle detection and path planning algorithms

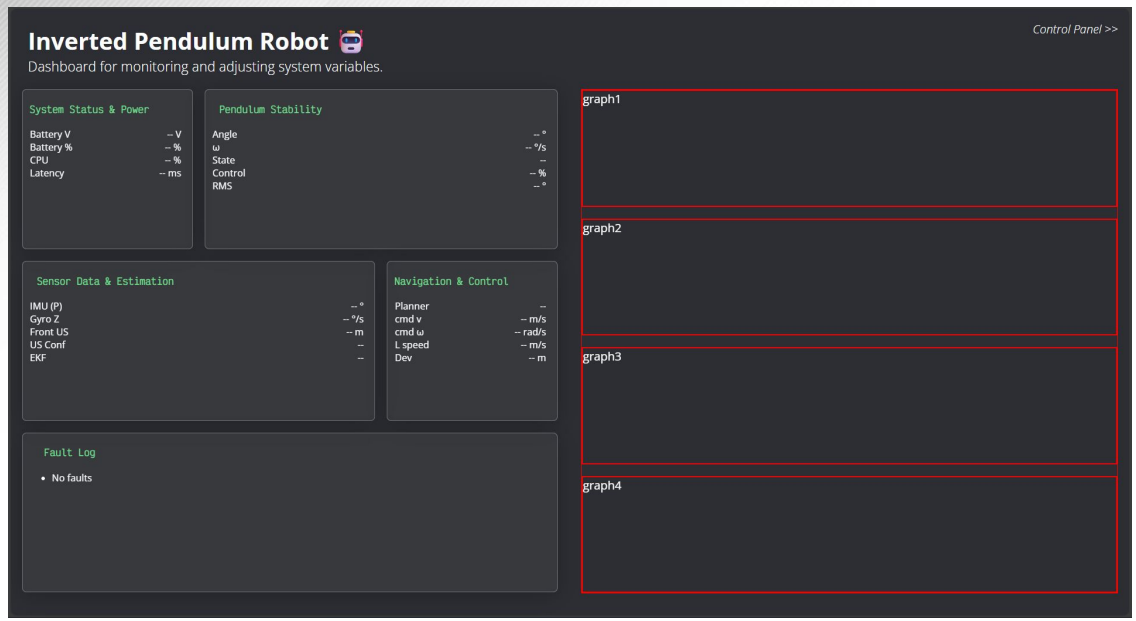
Continuing Work

- Optimize path planner for efficient replans/routing
- Improve IMU drift and sensor outlier filtering

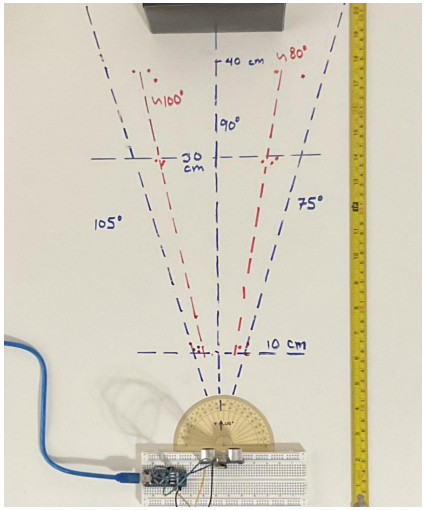
Planned for 404

- Integrate with MCU and controls subsystem
- Implement sensor fusion (EKF) to combine IMU and ultrasonic sensor data for closed-loop feedback system
- Optimize sensor placement and coverage mapping
- Reduce latency telemetry performance

Sensors, Navigation, and GUI



Web interface for sensor and navigation feedback



Sensor FOV Validation

ax_bias	ay_bias	az_bias	gx_bias	gy_bias	gz_bias
0.066178	0.149982	0.131714	0.656403	-0.51436	-0.69221
ax_noise	ay_noise	az_noise	gx_noise	gy_noise	gz_noise
0.001969	0.001814	0.001703	0.011253	0.01135	0.007771

IMU steady state mean and noise for accelerometer and gyroscope

Validation plan

Task	Specifications and Success Criteria	Status	Responsible Engineer(s)
Logic Calculation Rate	Validate through timing tests on MCU execution cycle; must complete calculations in < 50ms	Untested	Gage
Data Transfer Rates	Measure message latency with time logs; must be < 100 ms difference	Untested	Gage
Oscillator Precision	Validate clock stability through frequency tests; ensure deviation < 10ppm	Untested	Gage
Sonar Read Distance	Validate using logged ultrasonic data (CSV) and IMU; confirm ≥95% obstacle detection at distances ≥2 cm and ±3 cm accuracy within 0.2-1.0 m	Tested	Eric
Sensor Field-of-view Testing	Verify that the ultrasonic sensor coverage meets the ±15° design requirements	Tested	Eric
IMU Orientation Error	Verify that orientation measurements have ≤ 2° steady-state error and minimal drift	Tested	Eric
Telemetry/GUI Latency	Mean latency < 200 ms with minimal packet loss	Untested	Eric
Path Efficiency Ratio		Untested	Eric
Verify Input Power Delivery	Measure system voltage under load; confirm range is within 11.1-12.6V	Tested	Kaitlynn
Buck Converter Regulation	Record output voltage under different loads; verify 5V stability	Untested	Kaitlynn
LDO Validation	Measure regulated output with oscilloscope; confirm 3.3V stability	Untested	Kaitlynn
Protection Elements	Make sure the robot's power system is safe and protected from short circuits, overcurrent, or low battery conditions.	Untested	Kaitlynn
Control Algorithm Validation	Validate stability across various operating conditions and confirm tuned gains achieve target response time	Untested	Moustafa

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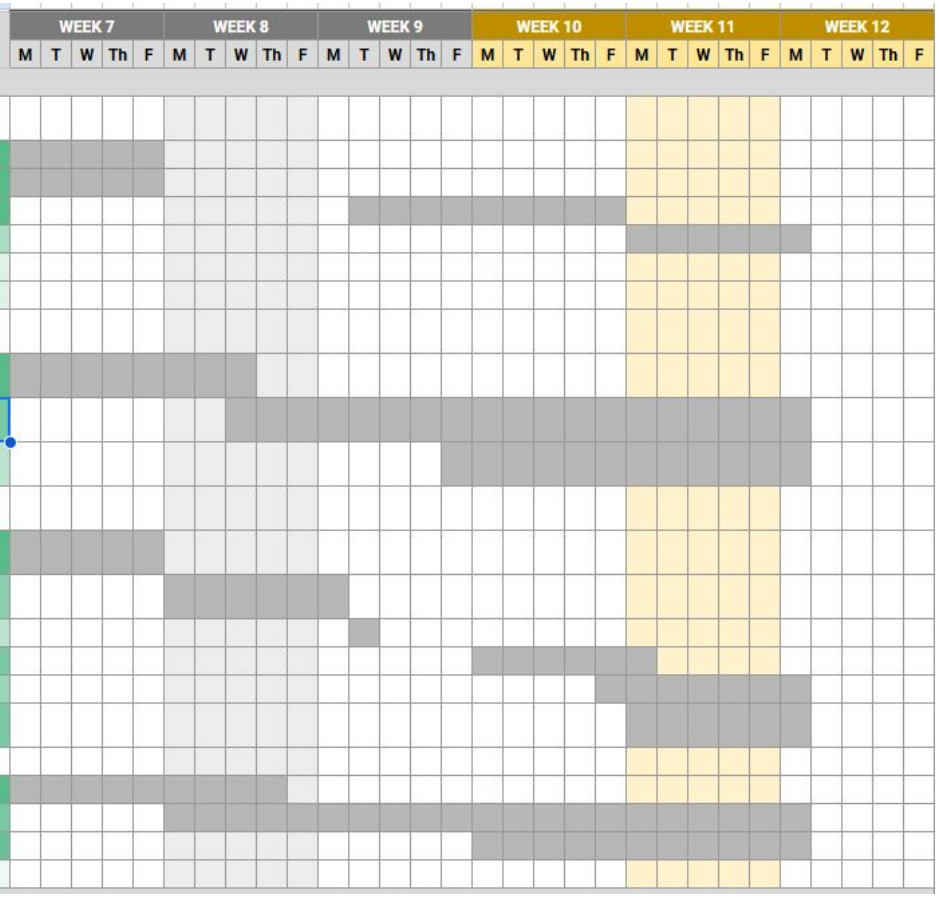
Execution Plan

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION	PCT OF TASK COMPLETE
3	Subsystem Development and Testing					
3.1	Power and mechanical subsystem construction	Kaitlynn L.				
	Design Linear Dropout Regulator PCB	Kaitlynn L.	10/6/25	10/10/25	4	100%
3.1.1	Design buck converter PCB	Kaitlynn L.	10/10/25	10/25/25	7	100%
3.1.3	Cut and join aluminum chassis	Kaitlynn L.	10/19/25	10/20/25	1	100%
3.1.4	Mount and install power electronics	Kaitlynn L.	11/20/25	11/29/25	9	50%
3.1.5	Mount and test pendulum	Kaitlynn L.	11/1/25	11/20/25	19	20%
3.1.6	Subsystem validation	Kaitlynn L.	11/24/25	11/27/25	3	20%
3.2	Controls subsystem construction	Moustafa K.				
3.2.1	Create LQR and PID control algorithm based of control model	Moustafa K.	10/06/25	10/15/25	9	100%
3.2.2	Code the control algorithm	Moustafa K.	10/15/25	10/27/25	12	80%
3.2.3	Subsystem validation	Moustafa K.	10/27/25	11/08/25	12	40%
3.3	Sensors & navigation subsystem contruction	Eric C.				
3.3.1	Setup sensor interfaces (I^2C communication)	Eric C.	10/06/25	10/12/25	6	100%
3.3.2	Test raw data; design filtering and fusion algorithms	Eric C.	10/13/25	10/26/25	6	70%
3.3.3	Test filtering algorithms with MATLAB	Eric C.	11/03/25	11/09/25	6	40%
3.3.4	Wifi implementation for MCU	Eric C.	10/27/25	11/03/25	7	80%
3.3.5	Develop path planning algorithm	Eric C.	10/31/25	11/05/25	5	60%
3.3.6	Validate sensors, filters, GUI; compare with metrics	Eric C.	11/17/25	11/23/25	6	80%
3.4	MCU subsystem construction	Gage F.				
3.4.1	MCU circuit design	Gage F.	10/06/25	10/15/25	9	100%
3.4.2	MCU interfacing, and protocols	Gage F.	10/06/25	10/15/25	9	80%
3.4.3	PCB Design	Gage F.	10/27/25	11/03/25	7	90%
3.4.5	Subsystem Validation	Gage F.	11/03/25	11/08/25	5	10%



Thank You!

Inverted Pendulum Robot 1

Any Questions?